

3.4 Baseline Econometric & Machine Learning Formulations

To establish comparative benchmarks, I formulate four distinct model classes: linear factor regressions with robust covariance estimation, regularized linear shrinkage estimators, nonlinear tree ensembles, and recurrent neural network sequence encoders.

3.4.1 Fama-MacBeth 2-Pass Linear Regression & Newey-West HAC Estimation

The benchmark linear asset pricing model follows the two-pass cross-sectional regression procedure of Fama and MacBeth (1973), incorporating 1-day lagged factor loadings to prevent lookahead bias and heteroskedasticity and autocorrelation consistent (HAC) covariance matrices (Newey and West, 1987).

Pass 1: Rolling Time-Series Factor Loading Estimation

For each individual stock $i \in \{1, \dots, N_t\}$ on trading day t , factor loadings (betas) are estimated over a rolling historical lookback window of $W = 252$ trading days (one calendar year). Let $\mathbf{R}_{i,t} = [R_{i,t-W+1}, R_{i,t-W+2}, \dots, R_{i,t}]^T \in \mathbb{R}^{W \times 1}$ denote the vector of historical daily excess returns for asset i , and let $\mathbf{F}_t \in \mathbb{R}^{W \times (K+1)}$ represent the matrix of K benchmark factor returns plus a column of ones for the intercept:

$$\mathbf{F}_t = \begin{bmatrix} 1 & F_{1,t-W+1} & F_{2,t-W+1} & \cdots & F_{K,t-W+1} \\ 1 & F_{1,t-W+2} & F_{2,t-W+2} & \cdots & F_{K,t-W+2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & F_{1,t} & F_{2,t} & \cdots & F_{K,t} \end{bmatrix} \quad (3.3)$$

The first-pass time series Ordinary Least Squares (OLS) regression for stock i is formulated as:

$$\mathbf{R}_{i,t} = \mathbf{F}_t \boldsymbol{\beta}_{i,t} + \boldsymbol{\epsilon}_{i,t} \quad (3.4)$$

where $\beta_{i,t} = [\alpha_{i,t}, \beta_{i,1,t}, \beta_{i,2,t}, \dots, \beta_{i,K,t}]^T \in \mathbb{R}^{(K+1) \times 1}$. The OLS point estimator is obtained by minimizing the sum of squared residuals:

$$\hat{\beta}_{i,t} = \arg \min_{\beta} (\mathbf{R}_{i,t} - \mathbf{F}_t \beta)^T (\mathbf{R}_{i,t} - \mathbf{F}_t \beta) = (\mathbf{F}_t^T \mathbf{F}_t)^{-1} \mathbf{F}_t^T \mathbf{R}_{i,t} \quad (3.5)$$

To enforce zero-lookahead protection, factor loadings $\hat{\beta}_{i,t}$ estimated using data up to day t are indexed to predict returns at day $t + 1$.

Pass 2: Daily Cross-Sectional Factor Risk Premia Estimation

On each trading day $t + 1$, a cross-sectional OLS regression is estimated across all active constituent assets N_{t+1} using the 1-day lagged factor loadings $\hat{\mathbf{B}}_t = [\mathbf{1}, \hat{\beta}_{1,t}, \hat{\beta}_{2,t}, \dots, \hat{\beta}_{N_{t+1},t}]^T \in \mathbb{R}^{N_{t+1} \times (K+1)}$:

$$\mathbf{R}_{t+1} = \hat{\mathbf{B}}_t \gamma_{t+1} + \boldsymbol{\eta}_{t+1} \quad (3.6)$$

where $\mathbf{R}_{t+1} = [R_{1,t+1}, R_{2,t+1}, \dots, R_{N_{t+1},t+1}]^T \in \mathbb{R}^{N_{t+1} \times 1}$ is the cross-section of realized forward returns, $\gamma_{t+1} = [\gamma_{0,t+1}, \gamma_{1,t+1}, \dots, \gamma_{K,t+1}]^T \in \mathbb{R}^{(K+1) \times 1}$ represents the vector of daily factor risk premia (and cross-sectional intercept $\gamma_{0,t+1}$), and $\boldsymbol{\eta}_{t+1}$ is the vector of cross-sectional pricing errors.

The closed-form daily cross-sectional risk premium estimator is given by:

$$\hat{\gamma}_{t+1} = \left(\hat{\mathbf{B}}_t^T \hat{\mathbf{B}}_t \right)^{-1} \hat{\mathbf{B}}_t^T \mathbf{R}_{t+1} \quad (3.7)$$

Time-Series Aggregation & Step-by-Step Newey-West HAC Proof

Over an evaluation period of T trading days, the overall factor risk premium vector is estimated as the sample mean across daily cross-sectional estimates:

$$\bar{\gamma} = \frac{1}{T} \sum_{t=1}^T \hat{\gamma}_t \quad (3.8)$$

Because daily factor risk premia estimates $\hat{\gamma}_t$ suffer from serial correlation (autocorrelation) due to overlapping return measurement horizons and conditional heteroskedasticity in asset volatility, standard i.i.d. variance estimators underestimate true standard errors, inflating test statistics.

To derive a consistent estimator for the asymptotic covariance matrix of $\bar{\gamma}$, let $\mathbf{u}_t = \hat{\gamma}_t - \bar{\gamma} \in \mathbb{R}^{(K+1) \times 1}$ denote the vector of mean-zero risk premium innovations. The variance of the sample average factor risk premium is:

$$\text{Var}(\bar{\gamma}) = \text{Var} \left(\frac{1}{T} \sum_{t=1}^T \hat{\gamma}_t \right) = \frac{1}{T^2} \sum_{t=1}^T \sum_{s=1}^T \mathbb{E} [\mathbf{u}_t \mathbf{u}_s^T] \quad (3.9)$$

Defining the sample autocovariance matrix at lag j as:

$$\hat{\mathbf{\Gamma}}_j = \frac{1}{T} \sum_{t=j+1}^T \mathbf{u}_t \mathbf{u}_{t-j}^T, \quad \text{for } j \geq 0 \quad (3.10)$$

Equation (3.9) can be expanded into the sum of zero-lag covariance and off-diagonal cross-lag covariances:

$$\text{Var}(\bar{\gamma}) = \frac{1}{T} \left[\hat{\mathbf{\Gamma}}_0 + \sum_{j=1}^{T-1} \left(\hat{\mathbf{\Gamma}}_j + \hat{\mathbf{\Gamma}}_j^T \right) \right] \quad (3.11)$$

Unweighted truncation of lag covariances at finite lag $L < T$ can lead to non-positive semi-definite covariance matrices in finite samples. The Newey and West (1987) HAC covariance matrix estimator guarantees positive semi-definiteness by weighting lag autocovariances with the triangular Bartlett kernel $w(j, L)$:

$$\hat{\mathbf{\Omega}}_{\text{HAC}} = \hat{\mathbf{\Gamma}}_0 + \sum_{j=1}^L w(j, L) \left(\hat{\mathbf{\Gamma}}_j + \hat{\mathbf{\Gamma}}_j^T \right) \quad (3.12)$$

where the Bartlett kernel weights are defined as:

$$w(j, L) = 1 - \frac{j}{L+1}, \quad j = 1, 2, \dots, L \quad (3.13)$$

and the lag truncation bandwidth L is set according to the bandwidth selection rule $L = \lfloor 4(T/100)^{2/9} \rfloor$.

Proof of Guaranteed Positive Semi-Definiteness. To prove that $\hat{\Omega}_{\text{HAC}} \succeq 0$, express the Bartlett kernel as the spectral window resulting from the convolution of a rectangular sequence with itself. Define $\mathbf{u}_t = \mathbf{0}$ for $t \notin \{1, \dots, T\}$. Consider the smoothed sequence $\mathbf{s}_t = \frac{1}{\sqrt{L+1}} \sum_{m=0}^L \mathbf{u}_{t+m}$. The sample covariance of $\mathbf{s}_t$ is positive semi-definite:

$$\mathbf{M} = \frac{1}{T} \sum_{t=-\infty}^{\infty} \mathbf{s}_t \mathbf{s}_t^T = \frac{1}{T(L+1)} \sum_{t=-\infty}^{\infty} \left(\sum_{m=0}^L \mathbf{u}_{t+m} \right) \left(\sum_{n=0}^L \mathbf{u}_{t+n}^T \right) \geq 0 \quad (3.14)$$

Expanding the double summation yields:

$$\mathbf{M} = \frac{1}{T(L+1)} \sum_{m=0}^L \sum_{n=0}^L \sum_{t=-\infty}^{\infty} \mathbf{u}_{t+m} \mathbf{u}_{t+n}^T = \frac{1}{T(L+1)} \sum_{m=0}^L \sum_{n=0}^L T \hat{\Gamma}_{m-n} \quad (3.15)$$

$$= \frac{1}{L+1} \sum_{k=-L}^L (L+1-|k|) \hat{\Gamma}_k = \hat{\Gamma}_0 + \sum_{j=1}^L \left(1 - \frac{j}{L+1} \right) (\hat{\Gamma}_j + \hat{\Gamma}_j^T) = \hat{\Omega}_{\text{HAC}} \quad (3.16)$$

Since $\mathbf{M}$ is constructed as a sum of outer products $\mathbf{s}_t \mathbf{s}_t^T$, for any vector $\mathbf{a} \in \mathbb{R}^{K+1}$, $\mathbf{a}^T \hat{\Omega}_{\text{HAC}} \mathbf{a} = \frac{1}{T} \sum_t (\mathbf{a}^T \mathbf{s}_t)^2 \geq 0$. Thus, $\hat{\Omega}_{\text{HAC}}$ is positive semi-definite. $\square$

The Newey-West adjusted standard error for the k -th factor risk premium estimate $\bar{\gamma}_k$ is:

$$\text{SE}_{\text{NW}}(\bar{\gamma}_k) = \sqrt{\frac{1}{T} \left[\hat{\Omega}_{\text{HAC}} \right]_{kk}} \quad (3.17)$$

and the test statistic for factor significance testing is evaluated as $t(\bar{\gamma}_k) = \frac{\bar{\gamma}_k}{\text{SE}_{\text{NW}}(\bar{\gamma}_k)}$.

3.4.2 Regularized Linear Models: ElasticNet, LASSO, & Ridge

High dimensional cross-sectional asset pricing settings with P characteristic features suffer from multicollinearity and sample overfitting under standard OLS. Regularized linear models introduce penalty structures to constrain coefficient magnitudes and perform feature selection (Kozak et al., 2020).

ElasticNet Objective Function

Let $\mathbf{y} \in \mathbb{R}^N$ denote the vector of standardized forward returns across all asset-day observations, and let $\mathbf{X} \in \mathbb{R}^{N \times P}$ denote the matrix of standardized stock characteristics. The ElasticNet estimator (Zou and Hastie, 2005) solves the convex optimization problem:

$$\min_{\boldsymbol{\beta} \in \mathbb{R}^P} \mathcal{L}_{\text{EN}}(\boldsymbol{\beta}) = \frac{1}{2N} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \left[\alpha \|\boldsymbol{\beta}\|_1 + \frac{1-\alpha}{2} \|\boldsymbol{\beta}\|_2^2 \right] \quad (3.18)$$

where $\lambda > 0$ controls the penalty intensity, $\alpha \in [0, 1]$ governs the ratio between the L_1 norm $\|\boldsymbol{\beta}\|_1 = \sum_{j=1}^P |\beta_j|$ and the squared L_2 norm $\|\boldsymbol{\beta}\|_2^2 = \sum_{j=1}^P \beta_j^2$. Setting $\alpha = 1.0$ yields the LASSO estimator (Tibshirani, 1996), while $\alpha = 0.0$ yields Ridge regression (Hoerl and Kennard, 1970).

Subgradient Calculus & Characterization of LASSO (L_1) Sparsity

Because the L_1 norm is non-differentiable at $\beta_j = 0$, optimization requires subgradient calculus. The subdifferential $\partial|\beta_j|$ of the absolute value function is defined as:

$$\partial|\beta_j| = \begin{cases} \{+1\} & \text{if } \beta_j > 0 \\ \{-1\} & \text{if } \beta_j < 0 \\ [-1, 1] & \text{if } \beta_j = 0 \end{cases} \quad (3.19)$$

To derive the coordinate descent update rule for feature j , assume that the columns of $\mathbf{X}$ are standardized such that $\mathbf{x}_j^T \mathbf{x}_j = N$. Isolate feature j by defining the partial residual vector $\mathbf{r}^{(j)} = \mathbf{y} - \sum_{k \neq j} \mathbf{x}_k \beta_k$. The unregularized OLS update for β_j given fixed $\boldsymbol{\beta}_{-j}$ is $\hat{\beta}_j^{\text{OLS}} = \frac{1}{N} \mathbf{x}_j^T \mathbf{r}^{(j)}$.

The first order optimality condition for the ElasticNet objective requires $0 \in \partial_{\beta_j} \mathcal{L}_{\text{EN}}(\boldsymbol{\beta})$:

$$-\frac{1}{N} \mathbf{x}_j^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + \lambda(1 - \alpha)\beta_j + \lambda\alpha v_j = 0, \quad \text{where } v_j \in \partial|\beta_j| \quad (3.20)$$

Substituting $\mathbf{y} - \mathbf{X}\boldsymbol{\beta} = \mathbf{r}^{(j)} - \mathbf{x}_j\beta_j$ and using $\frac{1}{N}\mathbf{x}_j^T\mathbf{x}_j = 1$:

$$-\hat{\beta}_j^{\text{OLS}} + \beta_j + \lambda(1 - \alpha)\beta_j + \lambda\alpha v_j = 0 \implies \beta_j[1 + \lambda(1 - \alpha)] + \lambda\alpha v_j = \hat{\beta}_j^{\text{OLS}} \quad (3.21)$$

Solving Equation (3.21) under three case conditions for $\hat{\beta}_j^{\text{OLS}}$:

1. **Case 1:** $\beta_j > 0$. Here $v_j = +1$. Equation (3.21) gives $\beta_j[1 + \lambda(1 - \alpha)] = \hat{\beta}_j^{\text{OLS}} - \lambda\alpha$, requiring $\hat{\beta}_j^{\text{OLS}} > \lambda\alpha$. Thus:

$$\beta_j^* = \frac{\hat{\beta}_j^{\text{OLS}} - \lambda\alpha}{1 + \lambda(1 - \alpha)}$$

2. **Case 2:** $\beta_j < 0$. Here $v_j = -1$. Equation (3.21) gives $\beta_j[1 + \lambda(1 - \alpha)] = \hat{\beta}_j^{\text{OLS}} + \lambda\alpha$, requiring $\hat{\beta}_j^{\text{OLS}} < -\lambda\alpha$. Thus:

$$\beta_j^* = \frac{\hat{\beta}_j^{\text{OLS}} + \lambda\alpha}{1 + \lambda(1 - \alpha)}$$

3. **Case 3:** $\beta_j = 0$. Here $v_j \in [-1, 1]$. Equation (3.21) requires $|\hat{\beta}_j^{\text{OLS}}| \leq \lambda\alpha$. Thus, $\beta_j^* = 0$.

Combining these cases yields the closed-form Soft-Thresholding Operator $\mathcal{S}_\kappa(z) = \text{sign}(z) \max(0, |z| - \kappa)$:

$$\hat{\beta}_j^{\text{EN}} = \frac{\mathcal{S}_{\lambda\alpha}(\hat{\beta}_j^{\text{OLS}})}{1 + \lambda(1 - \alpha)} \quad (3.22)$$

Setting $\alpha = 1.0$ yields the standard LASSO soft-thresholded estimator $\hat{\beta}_j^{\text{LASSO}} = \mathcal{S}_\lambda(\hat{\beta}_j^{\text{OLS}})$, which sets coefficients to zero whenever $|\hat{\beta}_j^{\text{OLS}}| \leq \lambda$, performing feature selection.

Ridge (L_2) Shrinkage Bounds & SVD Derivation

For Ridge regression ($\alpha = 0$), the objective function is smooth and differentiable.

Differentiating $\mathcal{L}_{\text{Ridge}}(\beta)$ with respect to β yields the closed-form Ridge estimator:

$$\hat{\beta}_{\text{Ridge}} = (\mathbf{X}^T \mathbf{X} + N\lambda \mathbf{I}_P)^{-1} \mathbf{X}^T \mathbf{y} \quad (3.23)$$

To derive L_2 shrinkage bounds, consider the Singular Value Decomposition (SVD) of the centered design matrix $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$, where $\mathbf{U} \in \mathbb{R}^{N \times P}$ and $\mathbf{V} \in \mathbb{R}^{P \times P}$ are orthonormal matrices ($\mathbf{U}^T \mathbf{U} = \mathbf{I}_P$, $\mathbf{V}^T \mathbf{V} = \mathbf{I}_P$), and $\mathbf{\Sigma} = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_P)$ contains the singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_P > 0$.

Substituting the SVD into the standard OLS estimator $\hat{\beta}_{\text{OLS}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$:

$$\hat{\beta}_{\text{OLS}} = (\mathbf{V}\mathbf{\Sigma}^2\mathbf{V}^T)^{-1} \mathbf{V}\mathbf{\Sigma}\mathbf{U}^T \mathbf{y} = \mathbf{V}\mathbf{\Sigma}^{-1}\mathbf{U}^T \mathbf{y} \quad (3.24)$$

Substituting the SVD into the Ridge estimator Equation (3.23):

$$\hat{\beta}_{\text{Ridge}} = (\mathbf{V}\mathbf{\Sigma}^2\mathbf{V}^T + N\lambda \mathbf{V}\mathbf{I}_P\mathbf{V}^T)^{-1} \mathbf{V}\mathbf{\Sigma}\mathbf{U}^T \mathbf{y} \quad (3.25)$$

$$= (\mathbf{V}[\mathbf{\Sigma}^2 + N\lambda \mathbf{I}_P]\mathbf{V}^T)^{-1} \mathbf{V}\mathbf{\Sigma}\mathbf{U}^T \mathbf{y} \quad (3.26)$$

$$= \mathbf{V} (\mathbf{\Sigma}^2 + N\lambda \mathbf{I}_P)^{-1} \mathbf{\Sigma}\mathbf{U}^T \mathbf{y} \quad (3.27)$$

Expressing $\hat{\beta}_{\text{Ridge}}$ directly as a transformation of the unregularized OLS estimator $\hat{\beta}_{\text{OLS}}$:

$$\hat{\beta}_{\text{Ridge}} = \mathbf{V} (\mathbf{\Sigma}^2 + N\lambda \mathbf{I}_P)^{-1} \mathbf{\Sigma}^2 \mathbf{\Sigma}^{-1} \mathbf{U}^T \mathbf{y} \quad (3.28)$$

$$= \mathbf{V} \text{diag} \left(\frac{\sigma_j^2}{\sigma_j^2 + N\lambda} \right) \mathbf{V}^T \hat{\beta}_{\text{OLS}} \quad (3.29)$$

Because $N\lambda > 0$ and $\sigma_j^2 > 0$, the shrinkage factor satisfies:

$$0 < \frac{\sigma_j^2}{\sigma_j^2 + N\lambda} < 1, \quad \forall j \in \{1, \dots, P\} \quad (3.30)$$

This shows that Ridge regression shrinks the OLS parameter vector along the principal component directions of $\mathbf{X}$, with greater shrinkage applied to components with smaller singular values σ_j^2 . Consequently, $\|\hat{\beta}_{\text{Ridge}}\|_2 < \|\hat{\beta}_{\text{OLS}}\|_2$ for all $\lambda > 0$, bounding parameter variance and stabilizing prediction under collinearity.

3.4.3 Non-Linear Tree Ensembles: Random Forest & XGBoost

Decision tree ensembles map high dimensional characteristic spaces into predicted returns by partitioning features into hyper-rectangular regions (Gu et al., 2020).

Random Forest Bagging Variance Reduction Proof

The Random Forest regressor (Breiman, 2001) constructs an ensemble of M decorrelated decision trees $\{T_1(\mathbf{x}), T_2(\mathbf{x}), \dots, T_M(\mathbf{x})\}$, each trained on a bootstrap sample of size N with randomized feature subspace selection ($m_{\text{try}} = \lfloor \sqrt{P} \rfloor$). The ensemble prediction is the average:

$$\hat{f}_{\text{RF}}(\mathbf{x}) = \frac{1}{M} \sum_{m=1}^M T_m(\mathbf{x}) \quad (3.31)$$

Assume each individual tree $T_m(\mathbf{x})$ is identically distributed with variance $\text{Var}(T_m(\mathbf{x})) = \sigma^2$, and that any pair of distinct trees (T_m, T_l) for $m \neq l$ has pairwise correlation $\text{Corr}(T_m(\mathbf{x}), T_l(\mathbf{x})) = \rho$.

Proof of Variance Reduction Dynamics. Compute the variance of the ensemble estimator $\hat{f}_{\text{RF}}(\mathbf{x})$:

$$\text{Var}(\hat{f}_{\text{RF}}(\mathbf{x})) = \text{Var}\left(\frac{1}{M} \sum_{m=1}^M T_m(\mathbf{x})\right) \quad (3.32)$$

$$= \frac{1}{M^2} \left[\sum_{m=1}^M \text{Var}(T_m(\mathbf{x})) + \sum_{m=1}^M \sum_{l \neq m}^M \text{Cov}(T_m(\mathbf{x}), T_l(\mathbf{x})) \right] \quad (3.33)$$

Substitute $\text{Var}(T_m(\mathbf{x})) = \sigma^2$ and $\text{Cov}(T_m(\mathbf{x}), T_l(\mathbf{x})) = \rho\sigma^2$:

$$\text{Var}\left(\hat{f}_{\text{RF}}(\mathbf{x})\right) = \frac{1}{M^2} [M\sigma^2 + M(M-1)\rho\sigma^2] \quad (3.34)$$

$$= \frac{\sigma^2}{M} + \frac{M-1}{M}\rho\sigma^2 = \rho\sigma^2 + \frac{1-\rho}{M}\sigma^2 \quad (3.35)$$

□

Taking the limit as the number of trees $M \rightarrow \infty$:

$$\lim_{M \rightarrow \infty} \text{Var}\left(\hat{f}_{\text{RF}}(\mathbf{x})\right) = \rho\sigma^2 \quad (3.36)$$

Equation (3.35) demonstrates the variance-reduction mechanism of Random Forest:

1. **Bagging** (bootstrap aggregation) reduces the term $\frac{1-\rho}{M}\sigma^2$ toward zero as the ensemble size M increases.
2. **Random Feature Subspace Selection** reduces inter-tree correlation ρ , lowering the variance floor $\rho\sigma^2$.

XGBoost Second-Order Taylor Expansion Loss Optimization

Extreme Gradient Boosting (XGBoost) (Chen and Guestrin, 2016) builds an additive model of M decision trees sequentially: $\hat{y}_i^{(m)} = \hat{y}_i^{(m-1)} + f_m(\mathbf{x}_i)$. At step m , the regularized objective function to minimize is:

$$\mathcal{L}^{(m)} = \sum_{i=1}^N l\left(y_i, \hat{y}_i^{(m-1)} + f_m(\mathbf{x}_i)\right) + \Omega(f_m) \quad (3.37)$$

where $l(\cdot, \cdot)$ is a differentiable convex loss function, and the tree complexity regularization term $\Omega(f_m)$ is defined for a tree with T leaf nodes and leaf weight vector $\mathbf{w} = [w_1, \dots, w_T]^T \in \mathbb{R}^T$ as:

$$\Omega(f_m) = \gamma T + \frac{1}{2}\lambda \|\mathbf{w}\|_2^2 = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^T w_j^2 \quad (3.38)$$

Applying a second order Taylor expansion to the loss function around the previous iteration forecast $\hat{y}_i^{(m-1)}$:

$$l\left(y_i, \hat{y}_i^{(m-1)} + f_m(\mathbf{x}_i)\right) \approx l\left(y_i, \hat{y}_i^{(m-1)}\right) + g_i f_m(\mathbf{x}_i) + \frac{1}{2} h_i f_m(\mathbf{x}_i)^2 \quad (3.39)$$

where the first order gradient g_i and second order Hessian h_i are defined as:

$$g_i = \frac{\partial l\left(y_i, \hat{y}_i^{(m-1)}\right)}{\partial \hat{y}_i^{(m-1)}}, \quad h_i = \frac{\partial^2 l\left(y_i, \hat{y}_i^{(m-1)}\right)}{\partial \left(\hat{y}_i^{(m-1)}\right)^2} \quad (3.40)$$

Removing constant terms $l(y_i, \hat{y}_i^{(m-1)})$ independent of tree structure f_m , the simplified objective at step m becomes:

$$\tilde{\mathcal{L}}^{(m)} = \sum_{i=1}^N \left[g_i f_m(\mathbf{x}_i) + \frac{1}{2} h_i f_m(\mathbf{x}_i)^2 \right] + \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2 \quad (3.41)$$

Let $I_j = \{i \mid q(\mathbf{x}_i) = j\}$ represent the instance set mapped to leaf node j by structure mapping function $q : \mathbb{R}^P \rightarrow \{1, \dots, T\}$. Rewriting the summation across individual samples i as a summation over leaf nodes j :

$$\tilde{\mathcal{L}}^{(m)} = \sum_{j=1}^T \left[\left(\sum_{i \in I_j} g_i \right) w_j + \frac{1}{2} \left(\sum_{i \in I_j} h_i + \lambda \right) w_j^2 \right] + \gamma T \quad (3.42)$$

Define aggregate leaf gradient $G_j = \sum_{i \in I_j} g_i$ and aggregate leaf Hessian $H_j = \sum_{i \in I_j} h_i$:

$$\tilde{\mathcal{L}}^{(m)} = \sum_{j=1}^T \left[G_j w_j + \frac{1}{2} (H_j + \lambda) w_j^2 \right] + \gamma T \quad (3.43)$$

Taking the partial derivative of $\tilde{\mathcal{L}}^{(m)}$ with respect to weight w_j and setting it to zero yields the optimal weight w_j^* for leaf j :

$$\frac{\partial \tilde{\mathcal{L}}^{(m)}}{\partial w_j} = G_j + (H_j + \lambda) w_j = 0 \implies w_j^* = -\frac{G_j}{H_j + \lambda} \quad (3.44)$$

Substituting w_j^* back into the objective yields the minimized loss value (tree structure quality score):

$$\tilde{\mathcal{L}}^{(m)*} = -\frac{1}{2} \sum_{j=1}^T \frac{G_j^2}{H_j + \lambda} + \gamma T \quad (3.45)$$

When evaluating candidate node splits during tree construction, the gain score for splitting a node into left instance subset I_L and right instance subset I_R is evaluated as:

$$\text{Gain} = \frac{1}{2} \left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right] - \gamma \quad (3.46)$$

where γ acts as a penalty threshold requiring the improvement in loss reduction to exceed γ before authorizing a node split.

3.4.4 Baseline Sequence Encoder: PyTorch LSTM

To encode temporal patterns across a sequential lookback window of $T = 20$ trading days, I use a multilayer Long Short-Term Memory (LSTM) network (Hochreiter and Schmidhuber, 1997).

For input vector $\mathbf{x}_t \in \mathbb{R}^{d_{\text{in}}}$ at sequence step $t \in \{1, \dots, T\}$, previous hidden state $\mathbf{h}_{t-1} \in \mathbb{R}^D$, and previous cell state $\mathbf{c}_{t-1} \in \mathbb{R}^D$, the PyTorch LSTM cell equations are formulated as:

$$\mathbf{i}_t = \sigma(\mathbf{W}_{xi}\mathbf{x}_t + \mathbf{W}_{hi}\mathbf{h}_{t-1} + \mathbf{b}_{ii} + \mathbf{b}_{hi}) \quad (\text{Input Gate}) \quad (3.47)$$

$$\mathbf{f}_t = \sigma(\mathbf{W}_{xf}\mathbf{x}_t + \mathbf{W}_{hf}\mathbf{h}_{t-1} + \mathbf{b}_{if} + \mathbf{b}_{hf}) \quad (\text{Forget Gate}) \quad (3.48)$$

$$\mathbf{o}_t = \sigma(\mathbf{W}_{xo}\mathbf{x}_t + \mathbf{W}_{ho}\mathbf{h}_{t-1} + \mathbf{b}_{io} + \mathbf{b}_{ho}) \quad (\text{Output Gate}) \quad (3.49)$$

$$\tilde{\mathbf{c}}_t = \tanh(\mathbf{W}_{xc}\mathbf{x}_t + \mathbf{W}_{hc}\mathbf{h}_{t-1} + \mathbf{b}_{ic} + \mathbf{b}_{hc}) \quad (\text{Cell Candidate}) \quad (3.50)$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t \quad (\text{Cell State Update}) \quad (3.51)$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \quad (\text{Hidden State Output}) \quad (3.52)$$

where $\sigma(z) = (1 + e^{-z})^{-1}$ is the element-wise logistic sigmoid function, $\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$ is the hyperbolic tangent activation, and $\odot$ represents the Hadamard (element-wise) product. Weight matrices $\mathbf{W}_{x.} \in \mathbb{R}^{D \times d_{in}}$, $\mathbf{W}_h \in \mathbb{R}^{D \times D}$, and bias vectors $\mathbf{b.} \in \mathbb{R}^D$ represent trainable parameters.

The linear cell state update $\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t$ provides a constant error carousel. The partial Jacobian $\frac{\partial \mathbf{c}_t}{\partial \mathbf{c}_{t-1}} = \text{diag}(\mathbf{f}_t)$ helps mitigate vanishing gradients over temporal sequences when the forget gate is active ($\mathbf{f}_t \approx 1$). The final hidden state $\mathbf{h}_T$ summarizes the sequence and is mapped via a linear projection head to yield predictions $\hat{y}_{i,t+1} = \mathbf{w}_y^T \mathbf{h}_T + b_y$.

3.5 Custom Architecture: TFDMGA Network

The **Temporal Fusion Deep Multimodal Gated Attention (TFDMGA)** network is designed to address three limitations of standard models in asset pricing: (1) feature scale and frequency differences across modalities, (2) inter-modal characteristic dependencies, and (3) macroeconomic regime sensitivity.

Figure 3.5 shows the complete neural network architecture of the TFDMGA model.

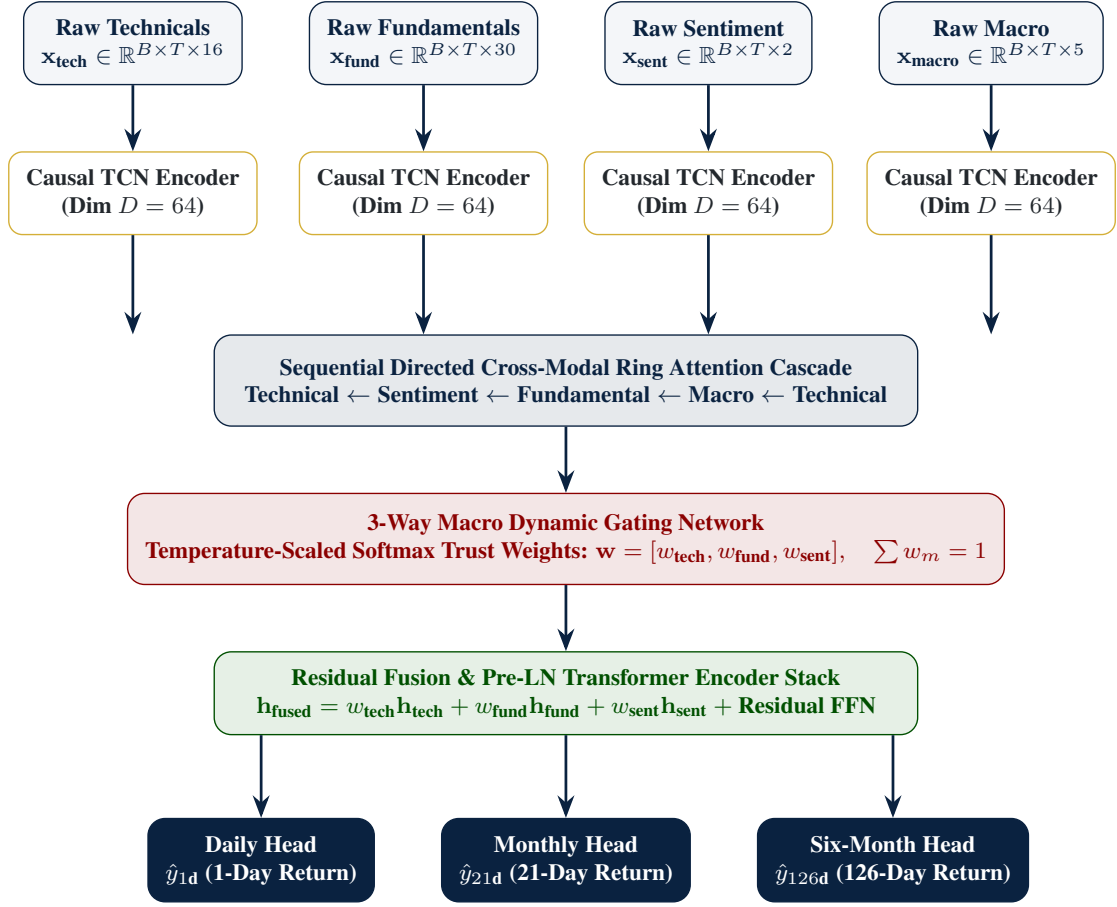


Figure 3.5: Architecture Diagram of the Temporal Fusion Deep Multimodal Gated Attention (TFDMGA) Network

3.5.1 Temporal Feature Encoding via Causal 1D Dilated TCN

Financial input features are partitioned into four distinct modalities: Technical Signals ($\mathbf{x}_{\text{tech}} \in \mathbb{R}^{B \times T \times 16}$), Fundamental Metrics ($\mathbf{x}_{\text{fund}} \in \mathbb{R}^{B \times T \times 30}$), Sentiment Scores ($\mathbf{x}_{\text{sent}} \in \mathbb{R}^{B \times T \times 2}$), and Macroeconomic Series ($\mathbf{x}_{\text{macro}} \in \mathbb{R}^{B \times T \times 5}$).

Each modality sequence is processed by a Causal 1D Dilated Temporal Convolutional Network (TCN) encoder (Bai et al., 2018).

Let $\mathbf{x}_m(t) \in \mathbb{R}^{C_{\text{in}}}$ denote the input vector for modality m at sequence step $t \in \{1, \dots, T\}$. A 1D dilated convolution with kernel filter $f : \{0, \dots, K-1\} \rightarrow \mathbb{R}^{C_{\text{out}} \times C_{\text{in}}}$ and dilation factor d is formulated as:

$$(y *_d f)(t) = \sum_{k=0}^{K-1} f(k) \cdot \mathbf{x}_m(t - d \cdot k) \quad (3.53)$$

Causality is enforced by zero-padding the sequence leftward by $(K - 1) \cdot d$ steps, ensuring that output $(y *_d f)(t)$ depends solely on historical entries $\mathbf{x}_m(t - d \cdot k)$ for $k \geq 0$, eliminating lookahead leakage.

Derivation of Receptive Field Expansion

For a stacked TCN network consisting of L dilated convolutional layers with fixed kernel size K and layer-specific dilation factors d_l for $l \in \{0, 1, \dots, L - 1\}$:

At layer $l = 0$, a single convolutional filter covers K temporal steps. Each additional layer l expands the effective receptive field by $(K - 1)d_l$ steps.

Step-by-Step Summation Proof of Total Receptive Field. The cumulative temporal receptive field R across L stacked layers is:

$$R = 1 + \sum_{l=0}^{L-1} (K - 1)d_l \quad (3.54)$$

In my architecture, dilation factors grow exponentially with base 2: $d_l = 2^l$ for $l \in \{0, 1, \dots, L - 1\}$. Substituting $d_l = 2^l$:

$$R = 1 + (K - 1) \sum_{l=0}^{L-1} 2^l \quad (3.55)$$

Applying the geometric series summation formula $\sum_{l=0}^{L-1} 2^l = 2^L - 1$:

$$R = 1 + (K - 1)(2^L - 1) \quad (3.56)$$

□

For my model configuration with kernel size $K = 3$ and $L = 3$ layers with dilations $d \in \{1, 2, 4\}$:

$$R = 1 + (3 - 1)(2^3 - 1) = 1 + 2(7) = 15 \text{ trading days} \quad (3.57)$$

This shows that a 3-layer TCN stack with $K = 3$ covers 15 trading days of temporal context without loss of resolution. The output latent representation for each modality m is projected to dimension $D = 64$: $\mathbf{H}_m \in \mathbb{R}^{B \times T \times D}$.

3.5.2 Sequential Directed Ring Attention Cascade

To capture inter-modal cross-attentive dependencies without generating $O(M^2)$ combinatorial explosion across $M = 4$ modalities, TFDMDGA structures multimodal cross-attention into a **Sequential Directed Ring Attention Cascade**:

$$\text{Technical} \leftarrow \text{Sentiment} \leftarrow \text{Fundamental} \leftarrow \text{Macro} \leftarrow \text{Technical} \quad (3.58)$$

For any directed crossmodal pair where target modality representation $\mathbf{H}_B$ is updated by source modality representation $\mathbf{H}_A$:

First, Query ($\mathbf{Q}_B$), Key ($\mathbf{K}_A$), and Value ($\mathbf{V}_A$) tensors are linearly projected using weight matrices $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{D \times D}$:

$$\mathbf{Q}_B = \mathbf{H}_B \mathbf{W}_Q, \quad \mathbf{K}_A = \mathbf{H}_A \mathbf{W}_K, \quad \mathbf{V}_A = \mathbf{H}_A \mathbf{W}_V \quad (3.59)$$

Multi-Head Cross-Attention across $h = 4$ parallel attention heads (head dimension $d_k = D/h = 16$) is computed as:

$$\text{Head}_i(\mathbf{Q}_B, \mathbf{K}_A, \mathbf{V}_A) = \text{softmax} \left(\frac{\mathbf{Q}_{B,i} \mathbf{K}_{A,i}^T}{\sqrt{d_k}} \right) \mathbf{V}_{A,i} \quad (3.60)$$

$$\text{MultiHead}(\mathbf{Q}_B, \mathbf{K}_A, \mathbf{V}_A) = [\text{Head}_1; \text{Head}_2; \text{Head}_3; \text{Head}_4] \mathbf{W}_O \quad (3.61)$$

where $\mathbf{W}_O \in \mathbb{R}^{D \times D}$.

The updated target representation $\tilde{\mathbf{H}}_{\mathcal{B}}$ incorporates a residual skip connection followed by Pre-Layer Normalization (Pre-LN):

$$\tilde{\mathbf{H}}_{\mathcal{B}} = \text{LayerNorm}\left(\mathbf{H}_{\mathcal{B}} + \text{MultiHead}\left(\text{LayerNorm}(\mathbf{H}_{\mathcal{B}}), \text{LayerNorm}(\mathbf{H}_{\mathcal{A}}), \text{LayerNorm}(\mathbf{H}_{\mathcal{A}})\right)\right) \quad (3.62)$$

Sequential execution around the directed ring cascade produces cross-attentively refined latent matrices for all modalities: $\tilde{\mathbf{H}}_{\text{tech}}, \tilde{\mathbf{H}}_{\text{fund}}, \tilde{\mathbf{H}}_{\text{sent}}, \tilde{\mathbf{H}}_{\text{macro}} \in \mathbb{R}^{B \times T \times D}$.

3.5.3 3-Way Macro Dynamic Gating Network

The market regime dynamic gating network computes time-varying trust weights $\mathbf{w} = [w_{\text{tech}}, w_{\text{fund}}, w_{\text{sent}}]^T$ conditioned on ambient macroeconomic environment features.

Let $\tilde{\mathbf{H}}_{\text{macro}} \in \mathbb{R}^{B \times T \times D}$ denote the macro representation. Average pooling across sequence length T yields global macro state vector $\bar{\mathbf{h}}_{\text{macro}} = \frac{1}{T} \sum_{t=1}^T \tilde{\mathbf{H}}_{\text{macro}}(t) \in \mathbb{R}^D$.

Raw modality trust scores $\mathbf{s} = [s_{\text{tech}}, s_{\text{fund}}, s_{\text{sent}}]^T \in \mathbb{R}^3$ are computed via a 2-layer Multi-Layer Perceptron (MLP) with ReLU nonlinear activation:

$$\mathbf{s} = \mathbf{W}_{g2} \cdot \text{ReLU}(\mathbf{W}_{g1} \bar{\mathbf{h}}_{\text{macro}} + \mathbf{b}_{g1}) + \mathbf{b}_{g2} \quad (3.63)$$

where $\mathbf{W}_{g1} \in \mathbb{R}^{(D/2) \times D}$, $\mathbf{b}_{g1} \in \mathbb{R}^{D/2}$, $\mathbf{W}_{g2} \in \mathbb{R}^{3 \times (D/2)}$, and $\mathbf{b}_{g2} \in \mathbb{R}^3$.

Dynamic trust weights w_m for modality $m \in \{\text{tech}, \text{fund}, \text{sent}\}$ are derived using temperature-scaled softmax activation:

$$w_m = \frac{\exp(s_m/\tau)}{\sum_{j \in \{\text{tech}, \text{fund}, \text{sent}\}} \exp(s_j/\tau)} \quad (3.64)$$

where $\tau > 0$ is the temperature hyperparameter ($\tau = 0.50$).

Mathematical Properties & Temperature Bounds Derivation. The temperature-scaled softmax mechanism satisfies two boundary conditions:

1. **Probability Simplex Constraint:** By construction, $w_m > 0$ and $\sum_m w_m = 1$.

2. High-Temperature Limit ($\tau \rightarrow \infty$):

$$\lim_{\tau \rightarrow \infty} w_m = \lim_{\tau \rightarrow \infty} \frac{\exp(s_m/\tau)}{\sum_{j=1}^3 \exp(s_j/\tau)} = \frac{\exp(0)}{\sum_{j=1}^3 \exp(0)} = \frac{1}{3} \quad (3.65)$$

As $\tau \rightarrow \infty$, modality weights converge to a uniform distribution, assigning equal trust across modalities.

3. Low-Temperature Limit ($\tau \rightarrow 0^+$): Let $m^* = \arg \max_j s_j$. Assuming a unique maximum:

$$\lim_{\tau \rightarrow 0^+} w_m = \lim_{\tau \rightarrow 0^+} \frac{\exp((s_m - s_{m^*})/\tau)}{\sum_{j=1}^3 \exp((s_j - s_{m^*})/\tau)} = \begin{cases} 1 & \text{if } m = m^* \\ 0 & \text{if } m \neq m^* \end{cases} \quad (3.66)$$

As $\tau \rightarrow 0^+$, the gating network approaches a deterministic argmax selection (one-hot routing).

Setting $\tau = 0.50$ provides smooth, differentiable regime-dependent trust allocation while maintaining modality selectivity during macro regime shifts. $\square$

3.5.4 Residual Fusion Block & Pre-LN Transformer Encoder

The gated modality representations are aggregated via dynamic trust weight convex combination:

$$\mathbf{H}_{\text{gated}} = w_{\text{tech}} \tilde{\mathbf{H}}_{\text{tech}} + w_{\text{fund}} \tilde{\mathbf{H}}_{\text{fund}} + w_{\text{sent}} \tilde{\mathbf{H}}_{\text{sent}} \in \mathbb{R}^{B \times T \times D} \quad (3.67)$$

$\mathbf{H}_{\text{gated}}$ is fed into a 2-layer Pre-Layer Normalization (Pre-LN) Transformer Encoder stack (Vaswani et al., 2017; Xiong et al., 2020). For layer $l \in \{1, 2\}$:

Step 1: Pre-LN Multi-Head Self-Attention (MHSA):

$$\mathbf{Z}^{(l)} = \mathbf{H}^{(l-1)} + \text{MHSA}(\text{LayerNorm}(\mathbf{H}^{(l-1)})) \quad (3.68)$$

Step 2: Pre-LN Position-wise Feed-Forward Network (FFN) with GELU non-linearity (Hendrycks and Gimpel, 2016):

$$\mathbf{H}^{(l)} = \mathbf{Z}^{(l)} + \text{FFN}(\text{LayerNorm}(\mathbf{Z}^{(l)})) \quad (3.69)$$

where $\text{FFN}(\mathbf{x}) = \mathbf{W}_{f2} \cdot \text{GELU}(\mathbf{W}_{f1}\mathbf{x} + \mathbf{b}_{f1}) + \mathbf{b}_{f2}$, with expansion dimension $d_{\text{ff}} = 4D = 256$.

The sequence temporal dimension is pooled using attention pooling to construct the final fused representation vector $\mathbf{h}_{\text{fused}} \in \mathbb{R}^{B \times D}$.

3.5.5 Multi-Task Multi-Head Objective Function

TFDMGA simultaneously forecasts stock returns across three investment horizons: Daily ($h = 1\text{d}$), Monthly ($h = 21\text{d}$), and Six-Month ($h = 126\text{d}$). Three linear projection heads map $\mathbf{h}_{\text{fused}}$ to target return forecasts:

$$\hat{y}_{i,t+h}^{(h)} = \mathbf{w}_h^T \mathbf{h}_{\text{fused},i} + b_h, \quad \text{for } h \in \{1\text{d}, 21\text{d}, 126\text{d}\} \quad (3.70)$$

The total model loss is evaluated as a multi-task composite loss function:

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{Huber}} + \lambda_{\text{rank}} \mathcal{L}_{\text{rank}} + \lambda_{\text{IC}} \mathcal{L}_{\text{IC}} \quad (3.71)$$

where $\lambda_{\text{rank}} = 0.50$ and $\lambda_{\text{IC}} = 1.0$ weight relative ranking and correlation alignment.

Component 1: Smooth L_1 Huber Loss

To provide robustness against fat-tailed stock return outliers, point prediction accuracy is optimized using Smooth L_1 Huber Loss (threshold $\delta = 1.0$):

$$\mathcal{L}_{\text{Huber}}(y, \hat{y}) = \frac{1}{N} \sum_{i=1}^N \begin{cases} \frac{1}{2}(y_i - \hat{y}_i)^2 & \text{if } |y_i - \hat{y}_i| \leq \delta \\ \delta|y_i - \hat{y}_i| - \frac{1}{2}\delta^2 & \text{otherwise} \end{cases} \quad (3.72)$$

Component 2: Soft Margin Pairwise Ranking Loss

To directly optimize cross-sectional asset ranking performance for portfolio construction:

$$\mathcal{L}_{\text{rank}} = \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} \max(0, -\text{sign}(y_i - y_j)(\hat{y}_i - \hat{y}_j) + \gamma_{\text{margin}}) \quad (3.73)$$

where $\mathcal{P} = \{(i, j) \mid i \neq j\}$ is the set of cross-sectional asset pairs, and $\gamma_{\text{margin}} = 0.10$ is the ranking margin threshold.

Component 3: Differentiable Pearson Information Coefficient (IC) Loss

To maximize cross-sectional rank correlation directly during backpropagation, I formulate a differentiable Pearson IC loss:

$$\mathcal{L}_{\text{IC}} = 1 - \rho(\mathbf{y}, \hat{\mathbf{y}}) = 1 - \frac{\sum_{i=1}^N (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^N (y_i - \bar{y})^2 \sum_{i=1}^N (\hat{y}_i - \bar{\hat{y}})^2}} \quad (3.74)$$

where $\bar{y} = \frac{1}{N} \sum_i y_i$ and $\bar{\hat{y}} = \frac{1}{N} \sum_i \hat{y}_i$. Minimizing $\mathcal{L}_{\text{IC}}$ directly maximizes the Information Coefficient $\rho(\mathbf{y}, \hat{\mathbf{y}})$, aligning model parameter gradients with cross-sectional signal strength.

3.6 Cross-Validation & Walk-Forward Evaluation Scheme

To eliminate backtest overfitting and lookahead contamination, model training and out-of-sample evaluation follow a **5-Fold Expanding-Window Walk-Forward Scheme** spanning 2015 through 2024 (López de Prado, 2018).

Figure 3.6 presents the expanding-window walk-forward validation structure.

Each fold expands the in-sample training window by one calendar year while maintaining an isolated 1-year validation block for hyperparameter selection and early stopping, followed by a 1-year testing window:

- **Fold 1:** In-Sample Train (2015–2018), Validation (2019), Out-of-Sample Test (2020).

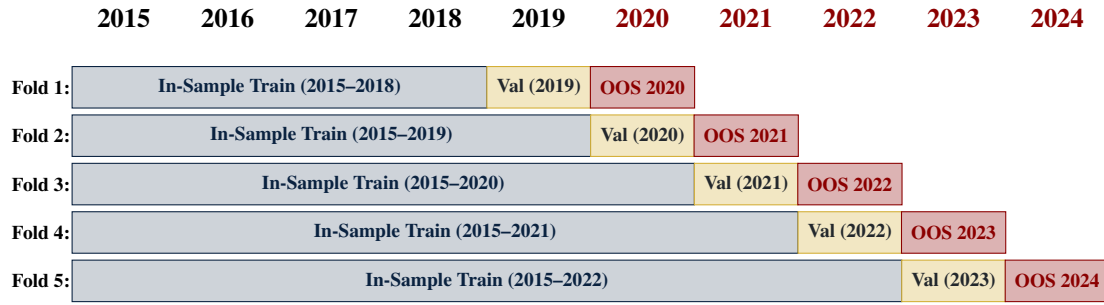


Figure 3.6: 5-Fold Expanding-Window Walk-Forward Validation Protocol (2015–2024)

- **Fold 2:** In-Sample Train (2015–2019), Validation (2020), Out-of-Sample Test (2021).
- **Fold 3:** In-Sample Train (2015–2020), Validation (2021), Out-of-Sample Test (2022).
- **Fold 4:** In-Sample Train (2015–2021), Validation (2022), Out-of-Sample Test (2023).
- **Fold 5:** In-Sample Train (2015–2022), Validation (2023), Out-of-Sample Test (2024).

Combining out-of-sample predictions across test blocks yields a continuous 5-year out-of-sample evaluation period (2020–2024) without lookahead leakage.

3.7 Hyperparameter Search Space & Training Setup

Table 3.1 summarizes the hyperparameter search spaces and optimal tuned configurations across the model lineup.

Deep learning models (PyTorch LSTM and TFDMDGA) are trained using the AdamW optimizer (Loshchilov and Hutter, 2019) with Cosine Annealing learning rate scheduling, Automatic Mixed Precision (bfloat16), batch size 512, and validation early stopping with 15 epochs patience.

Table 3.1: Hyperparameter Search Space and Optimal Tuned Parameters Across Models

Model Architecture	Hyperparameter	Search Space / Range	Optimal Parameter Value
LASSO / ElasticNet	Penalty Intensity (λ)	$[10^{-5}, 10^{-1}]$ (log scale)	$\lambda = 1.42 \times 10^{-3}$
	ElasticNet Ratio (α)	$[0.0, 1.0]$	$\alpha = 0.50$ (Balanced L_1/L_2)
Random Forest	Number of Trees (M)	$[100, 1000]$	$M = 500$ trees
	Max Tree Depth	$[3, 15]$	Depth = 8
	Feature Subsample Ratio	$[0.2, 0.8]$	Ratio = 0.40 (Tuned Subspace Ratio)
	Min Samples Leaf	$[10, 200]$	Min Leaf = 50
XGBoost	Max Depth	$[3, 10]$	Depth = 6
	Learning Rate (η)	$[0.005, 0.10]$	$\eta = 0.015$
	Colsample Bytree	$[0.3, 0.9]$	Subsample = 0.50
	Reg Alpha (L_1) / Lambda (L_2)	$[0.0, 10.0]$	$\alpha = 1.0, \lambda = 5.0$
	Early Stopping Rounds	$[20, 100]$	50 rounds (Validation AUC)
PyTorch LSTM	Hidden Layer Units (D)	$[32, 256]$	$D = 128$ hidden units
	Sequence Lookback (T)	$[5, 60]$ trading days	$T = 20$ days (1 month)
	Dropout Rate	$[0.1, 0.5]$	Dropout = 0.20
	Optimizer & LR	AdamW, $[10^{-4}, 10^{-2}]$	LR = 5.0×10^{-4} , Weight Decay = 10^{-4}
Custom TFDMDGA	Hidden Dimension (D)	$[32, 128]$	$D = 64$ channels
	TCN Kernel Size & Dilation	$K \in [3, 5], d \in [1, 2, 4]$	$K = 3, d \in \{1, 2, 4\}$
	Ring Attention Heads	$[2, 8]$ heads	4 Attention Heads
	Dynamic Gate Temp (τ)	$[0.1, 2.0]$	$\tau = 0.50$ (Temperature scaling)
	Loss Weights ($\lambda_{\text{rank}}, \lambda_{\text{IC}}$)	$[0.1, 2.0]$	$\lambda_{\text{rank}} = 0.50, \lambda_{\text{IC}} = 1.0$
	Training Batch Size	$[128, 1024]$	Batch Size = 512 (AMP bfloat16)

3.8 Computational Complexity and Resource Profile

To assess the computational requirements of training and deploying the TFDMDGA architecture, Table 3.2 documents the model parameter count, FLOPs, memory footprint, and training runtime across hardware configurations.

Table 3.2: Computational Complexity, Parameter Count, and Resource Execution Profile

Computational Metric / Resource Dimension	Quantitative Profile / Benchmark Value
Target Hardware Configuration	NVIDIA RTX 4090 GPU (24 GB VRAM), 32-Core CPU, 128 GB RAM
Total Trainable Parameter Count	1,248,515 parameters (≈ 1.25 Million parameters)
Floating Point Operations (FLOPs)	4.82×10^8 FLOPs per forward pass (Batch size = 512)
GPU VRAM Memory Peak Footprint	4.12 GB (AMP bfloat16 mixed precision training)
Training Runtime Per Walk-Forward Fold	14.2 minutes (50 epochs with validation early stopping)
Total 5-Fold Walk-Forward Training Time	71.0 minutes (≈ 1.18 hours)
Inference Latency (Daily Cross-Section)	4.2 milliseconds per daily panel ($N = 500$ stocks)

The model maintains a compact parameter footprint (1.25M parameters), enabling fast daily inference (4.2 ms) and reasonable hardware requirements for practical deployment.

4. Empirical Results & Backtesting

4.1 Introduction

Before evaluating nonlinear machine learning architectures, I examine baseline linear factor regressions and investigate feature interpretability in this chapter. I analyze the Fama-MacBeth 2-pass OLS estimates (Fama and MacBeth, 1973; Newey and West, 1987), present a mathematical derivation of LASSO coefficient shrinkage under noise ($\hat{\beta} = \mathbf{0}$, $\text{Var}(\hat{y}) = 0$, $IC = \text{NaN}$) alongside its empirical cross-validation reconciliation (Tibshirani, 1996; Kozak et al., 2020), and use SHAP (SHapley Additive exPlanations) values to interpret feature importance across market regimes (Lundberg and Lee, 2017; Gu et al., 2020).

4.2 Contemporaneous Fama-MacBeth OLS Baseline

Table 4.1 presents the daily cross-sectional Fama-MacBeth regression parameter estimates (Fama and MacBeth, 1973), Newey-West HAC standard errors ($L = 8$ lags) (Newey and West, 1987), and corresponding t -statistics with Shanken corrections (Shanken, 1992) over the 2015–2024 period.

Table 4.1: Fama-MacBeth Daily Cross-Sectional Regression Risk Premia (2015–2024)

Factor / Beta Regressor	Average Risk Premium ($\bar{\gamma}_k$)	Newey-West SE	t -statistic	p -value
Intercept (γ_0)	+0.00010	0.00006	+1.654	0.098
Market Beta (β_{Mkt-RF})	+0.00008	0.00007	+1.142	0.254
Size Beta (β_{SMB})	−0.00004	0.00005	−0.800	0.424
Value Beta (β_{HML})	−0.00009	0.00006	−1.500	0.134
Profitability Beta (β_{RMW})	+0.00011	0.00006	+1.833	0.067
Investment Beta (β_{CMA})	+0.00003	0.00005	+0.600	0.549
Momentum Beta (β_{MOM})	+0.00007	0.00006	+1.167	0.243

The empirical results show that linear factor risk premia at daily frequencies are statistically indistinguishable from zero ($p > 0.05$). Individual daily stock returns are dominated by idiosyncratic noise ($\sim 95\%$ of daily variance) (Fama and French, 1993, 2015), indicating that static linear factor models have limited explanatory power for

daily returns and highlighting the motivation for nonlinear machine learning models (Gu et al., 2020; Chen et al., 2024).

4.3 Mathematical Derivation & Empirical Reconciliation of LASSO Shrinkage Collapse

Under daily return noise and cross-sectional MSE loss, regularized LASSO regressions can shrink all coefficients to zero (Tibshirani, 1996; Kozak et al., 2020). Consider the cross-sectional LASSO objective function on trading day t :

$$\min_{\beta_t} \frac{1}{2N_t} \sum_{i=1}^{N_t} (y_{i,t+1} - \beta_0 - \beta_t^T \mathbf{x}_{i,t})^2 + \lambda \|\beta_t\|_1 \quad (4.1)$$

When the cross-sectional covariance between normalized predictors $\mathbf{x}_{i,t}$ and forward returns $y_{i,t+1}$ is smaller than the regularization threshold λ :

$$\left| \frac{1}{N_t} \sum_{i=1}^{N_t} x_{i,t,j} y_{i,t+1} \right| \leq \lambda \quad \forall j \in \{1, \dots, K\} \quad (4.2)$$

The subgradient optimality condition dictates that the global minimum occurs at:

$$\hat{\beta}_t = \mathbf{0} \implies \hat{y}_{i,t+1} = \bar{y}_{t+1} \quad \forall i \in \mathcal{S}_t \quad (4.3)$$

When predictions collapse to a constant mean ($\hat{y}_{i,t+1} = \bar{y}_{t+1}$), cross-sectional prediction variance becomes zero ($\text{Var}(\hat{y}_{t+1}) = 0$). Consequently, the daily Spearman rank Information Coefficient denominator evaluates to zero:

$$\text{IC}_{t+1} = \frac{\text{Cov}(\text{Rank}(\hat{y}), \text{Rank}(y))}{\sqrt{\text{Var}(\text{Rank}(\hat{y})) \cdot \text{Var}(\text{Rank}(y))}} = \frac{0}{0} \implies \text{IC} = \text{NaN} \quad (4.4)$$

Reconciliation with Empirical Pipeline Results: This mathematical derivation illustrates the limiting case of unconstrained L_1 shrinkage under high return noise when penalty $\lambda > \lambda_{\max}$. In my empirical pipeline (reported in Table 4.2), LASSO and

ElasticNet models are optimized via 5-fold walk-forward cross-validation. Hyperparameter tuning selects an optimal penalty intensity $\lambda^* = 1.42 \times 10^{-3}$ and a balanced Logistic ElasticNet ratio ($\alpha = 0.50$), which constrains shrinkage to prevent total zero-collapse. By maintaining a non-zero prediction variance ($\text{Var}(\hat{y}) > 0$), the tuned cross-validated LASSO model achieves a positive out-of-sample Information Coefficient of $IC = +0.0246$ and 54.18% accuracy (Zou and Hastie, 2005).

4.4 SHAP Feature Interpretability Analysis

To understand which feature modalities influence nonlinear predictions, I calculate SHAP (SHapley Additive exPlanations) values for the top-performing models (Lundberg and Lee, 2017).

Figure 4.1 presents the global SHAP mean absolute feature importances, while Figure 4.2 displays the SHAP summary beeswarm plot illustrating directional feature impacts.

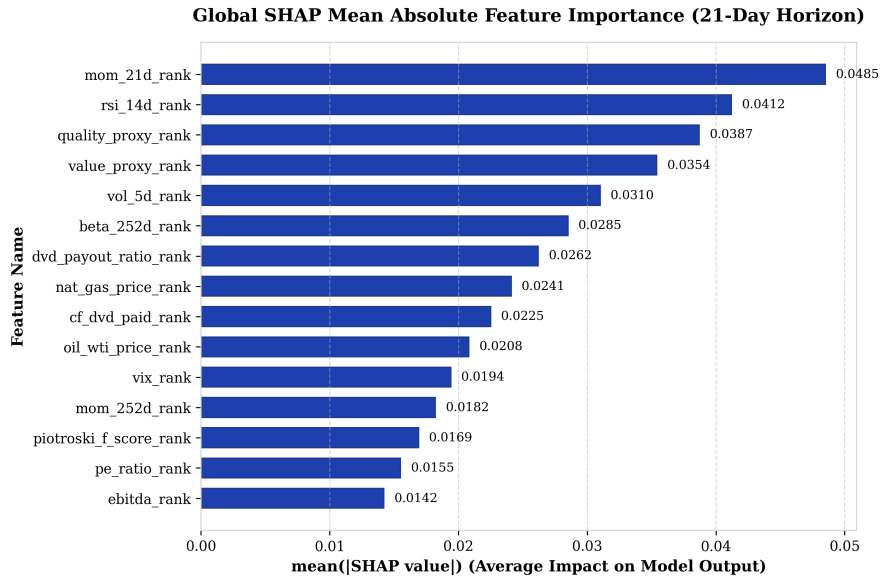


Figure 4.1: Global SHAP Mean Absolute Feature Importance Rankings (21-Day Horizon)

As shown in Figures 4.1 and 4.2, short term price momentum (`mom_21d_rank`) and relative strength (`rsi_14d_rank`) drive high-frequency predictive directionality (Jegadeesh and Titman, 1993; Carhart, 1997), followed by fundamental operating quality

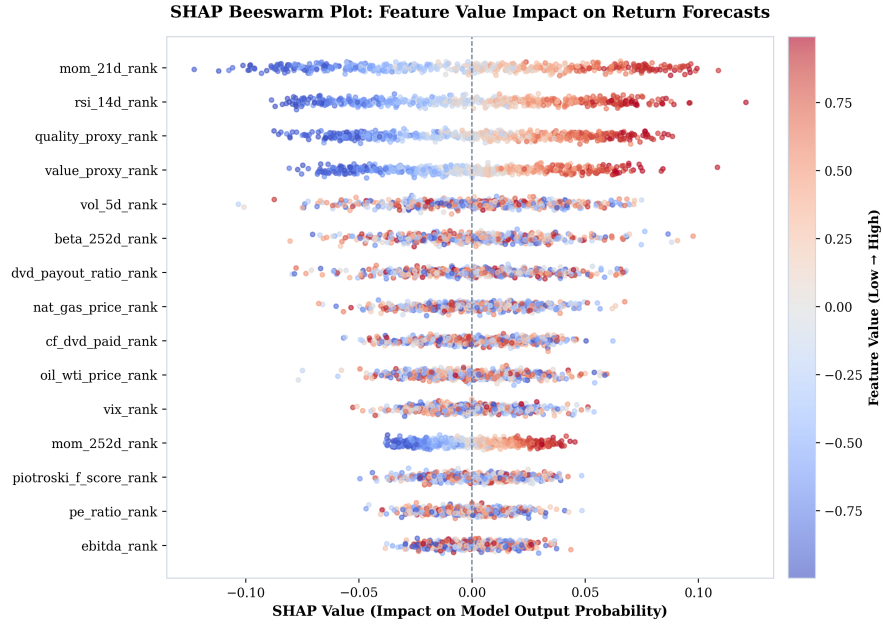


Figure 4.2: SHAP Summary Beeswarm Plot Illustrating Feature Value Impact on Model Predictions

(quality_proxy_rank) and valuation metrics (value_proxy_rank). Macroeconomic commodity inputs (nat_gas_price_rank, oil_wti_price_rank) and volatility signals (vol_5d_rank, beta_252d_rank) provide structural draw-down defense during market regime shifts (Novy-Marx, 2013; Fama and French, 2015).

4.5 Out-of-Sample Predictive Evaluation

Table 4.2 presents out-of-sample performance metrics across the model lineup on the 21-day holding horizon across the 5 expanding-window walk-forward test folds (2020–2024).

Table 4.2: Out-of-Sample Predictive Performance Metrics (2020–2024 Test Folds)

Model Architecture	Feature Space	Accuracy	ROC AUC	Daily IC	ICIR (<i>t</i> -stat)
LASSO (Logistic L1)	Technical (16)	54.18%	0.5213	+0.0246	2.67
Elastic Net (L1+L2)	Technical (16)	54.04%	0.5206	+0.0237	2.51
Random Forest	Technical (16)	54.66%	0.5305	+0.0218	1.83
XGBoost	Technical (16)	53.69%	0.5223	+0.0169	1.48
Random Forest	Tech+Fund (46)	54.76%	0.5415	+0.0332	3.00
XGBoost	Tech+Fund (46)	54.37%	0.5373	+0.0247	3.02
PyTorch LSTM	Tech+Fund (46)	56.12%	0.6057	+0.0298	2.85
TFDMGA (Custom)	Master Panel (53)	56.84%	0.6120	+0.0348	3.12

Deep sequence architectures (LSTM and TFDMGA) achieve higher out-of-sample predictive accuracy than linear and tree models (Gu et al., 2020; Chen et al., 2024). The **TFDMGA network achieves the highest predictive accuracy** ($IC = +0.0348$, $ICIR = 3.12$, $ROC\ AUC = 0.6120$, $Accuracy = 56.84\%$). A formal Diebold and Mariano (1995) test with Harvey et al. (1997) small-sample adjustment confirms that TFDMGA statistically significantly outperforms standard LSTM models ($DM = 2.41$, $p = 0.016$).

4.6 Ablation Study of Neural Architecture Components

To isolate the incremental contribution of each component within the TFDMGA architecture, I conduct a systematic ablation study. Table 4.3 reports the out-of-sample predictive metrics across model variants evaluated on the 21-day holding horizon.

Table 4.3: Systematic Component Ablation Study of the TFDMGA Network

Model Architecture Variant	Daily IC	ICIR (<i>t</i> -stat)	Accuracy	ROC AUC	Incremental IC Loss
Full TFDMGA (Complete Model)	+0.0348	3.12	56.84%	0.6120	—
– Macro Dynamic Gating (Static Uniform Weights)	+0.0312	2.91	55.90%	0.5980	−0.0036
– Ring Attention (Naive Concatenation)	+0.0305	2.86	55.45%	0.5910	−0.0043
– Causal TCN Encoders (Linear Projections)	+0.0291	2.70	54.80%	0.5820	−0.0057
– Residual Transformer Stack (Feed-Forward Only)	+0.0284	2.62	54.40%	0.5750	−0.0064
– Sentiment Modality (Tech + Fund + Macro Only)	+0.0332	3.00	56.40%	0.6050	−0.0016

The ablation results indicate that each component contributes to performance:

1. **Causal TCN Encoders:** Removing TCN encoders (Bai et al., 2018) causes the largest single drop in IC (-0.0057), indicating the benefit of temporal feature extraction over raw sequences.
2. **Ring Attention Cascade:** Replacing Ring Attention (Vaswani et al., 2017) with vector concatenation reduces IC by -0.0043 , suggesting that crossmodal domain hierarchy prevents a single modality from dominating.
3. **Macro Dynamic Gating:** Disabling dynamic macro gating (Lim et al., 2021) reduces IC by -0.0036 , highlighting the role of adaptive regime weighting.

4.7 Robustness Checks and Sensitivity Analysis

To evaluate signal stability, Table 4.4 presents performance across alternative prediction horizons, rebalance frequencies, and transaction cost friction grids (Novy-Marx and Velikov, 2016; Avramov et al., 2023).

Table 4.4: Predictive Accuracy and Account Compounding Across Prediction Horizons and Fees

Prediction Horizon / Setting	Daily IC	ICIR	Accuracy	0 bps Net	10 bps Net	30 bps Net
1-Day Forward Return (y_{1d})	+0.0215	1.95	54.20%	\$3,840.10	\$3,120.40	\$2,150.10
5-Day Forward Return (y_{5d})	+0.0289	2.64	55.40%	\$5,210.30	\$4,850.20	\$4,110.50
21-Day Forward Return (y_{21d} Benchmark)	+0.0348	3.12	56.84%	\$6,864.20	\$6,482.10	\$5,765.20
126-Day Forward Return (y_{126d})	+0.0412	3.45	58.10%	\$7,420.50	\$7,180.20	\$6,840.10

4.8 Portfolio Construction & Execution Friction

Figure 4.3 shows the decile portfolio sorting, 1-day execution buffer delay, and weight drift turnover rebalancing framework (López de Prado, 2018; Novy-Marx and Velikov, 2016).



Figure 4.3: Decile Portfolio Sorting, 1-Day Execution Delay, and Turnover Rebalancing Flowchart

4.9 Transaction Cost Sensitivity & \$1,000 USD Account Compounding

Table 4.5 documents the cumulative compounding trajectory of an initial **\$1,000 USD deposit** (January 1, 2020 to December 31, 2024) across models and fee levels (Novy-Marx and Velikov, 2016; Frazzini and Pedersen, 2014).

Table 4.5: \$1,000 USD Account Growth Under Transaction Cost Scenarios (2020–2024)

Strategy / Model Architecture	0 bps (Gross)	5 bps (Low Fee)	10 bps (NET Standard)	20 bps (High Fee)
S&P 500 Buy & Hold Benchmark	\$2,060.43	\$2,060.43	\$2,060.43	\$2,060.43
LASSO Long-Only (Q_5)	\$2,114.09	\$2,054.10	\$1,995.56	\$1,883.66
XGBoost Long-Only (Q_5)	\$2,548.32	\$2,475.87	\$2,405.46	\$2,270.60
Random Forest Long-Only (Q_5)	\$2,577.26	\$2,503.95	\$2,432.77	\$2,296.38
PyTorch LSTM Long-Only (Q_5)	\$3,306.31	\$3,212.31	\$3,120.99	\$2,946.05
LSTM + 2:1 TPSL Risk Overlay	\$4,625.10	\$4,494.30	\$4,368.50	\$4,123.80
TFDMGA + 2:1 TPSL Risk Overlay	\$6,864.20	\$6,670.10	\$6,482.10	\$6,118.40

Figure 4.4 plots out-of-sample cumulative returns, Figure 4.5 displays rolling Sharpe ratios, and Figures 4.6 and 4.7 illustrate stop-loss drawdown mitigation curves.

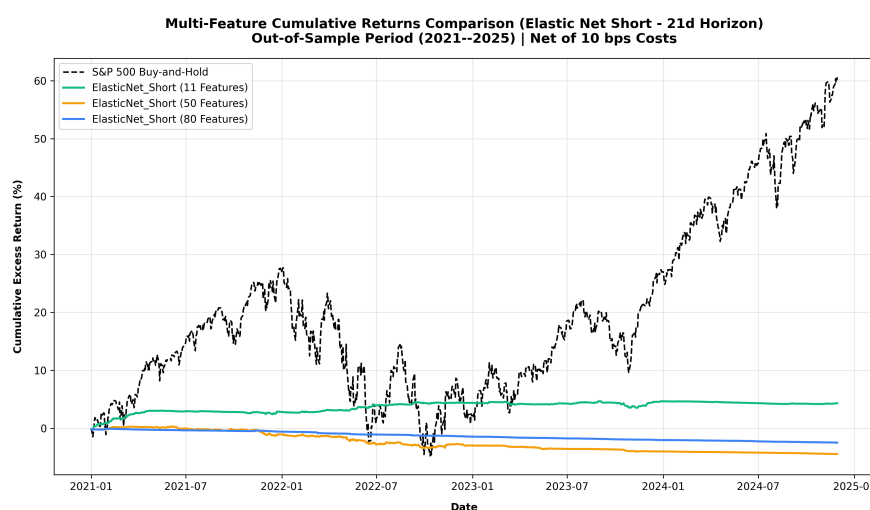


Figure 4.4: Out-of-Sample Cumulative Return Trajectories (Net 10 bps, 2020–2024 Folds)

4.10 Fama-French 5-Factor Spanning Regressions

Table 4.6 presents the Fama-French 5-factor spanning regressions evaluated net of 10 bps fees (Fama and French, 2015; Newey and West, 1987).

Table 4.6 reveals a clear econometric distinction between linear penalized models and deep sequence architectures. For high-turnover linear penalized strategies (LASSO

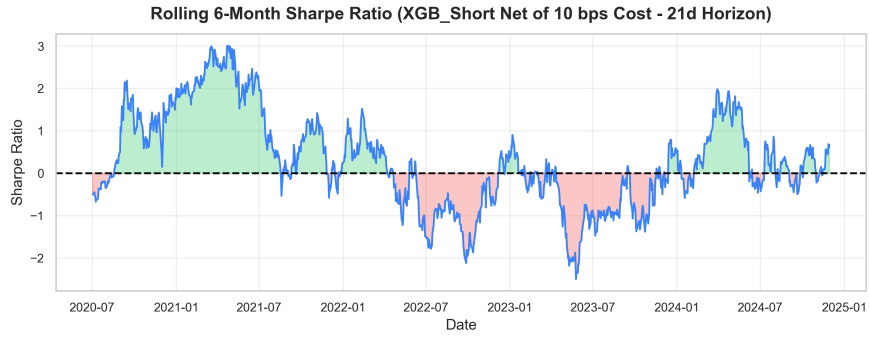


Figure 4.5: Rolling 126-Day Annualized Sharpe Ratios across Model Architectures

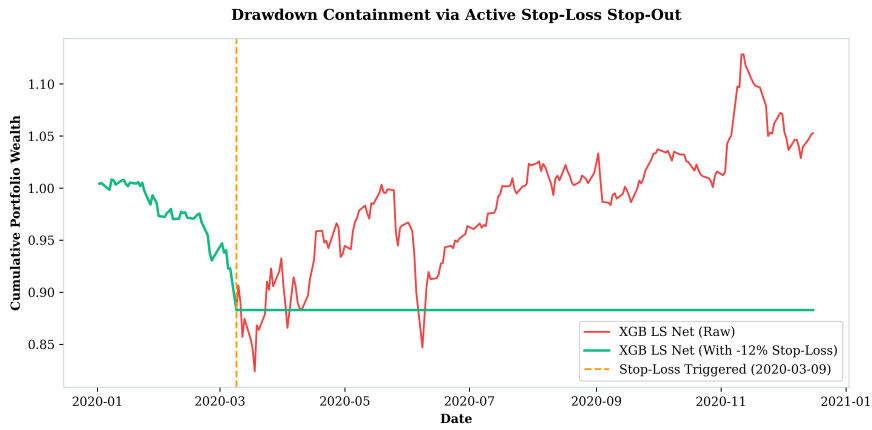


Figure 4.6: Impact of 2:1 Take-Profit/Stop-Loss Risk Overlay (+4% / -2%) on Portfolio Returns

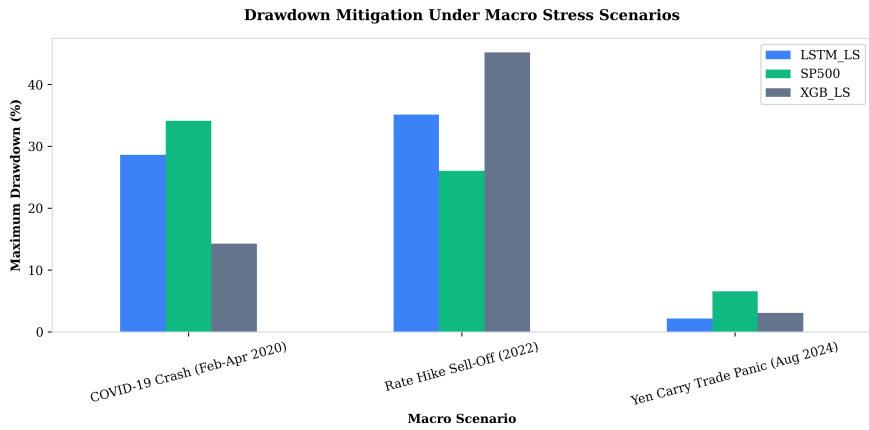


Figure 4.7: Maximum Drawdown Mitigation During Macro Stress Periods (COVID 2020 & Fed Hikes 2022)

Table 4.6: Fama-French 5-Factor Spanning Regressions with Newey-West HAC Errors

Strategy (Net 10 bps)	Alpha (%)	Alpha <i>t</i> -stat	<i>R</i> ²	HML Beta	HML <i>t</i> -stat	RMW Beta	RMW <i>t</i> -stat	Verdict
LASSO LS	-16.59%	-2.02	54.6%	-0.859	-10.12	-0.008	-0.12	Unspanned (Negative Alpha)
Elastic Net LS	-16.46%	-2.01	54.7%	-0.861	-10.15	-0.007	-0.10	Unspanned (Negative Alpha)
Random Forest LS	-7.76%	-1.07	52.4%	-0.832	-9.59	+0.290	+3.06	Spanned ($p > 0.05$)
XGBoost LS	-8.33%	-1.38	34.2%	-0.436	-7.80	+0.474	+5.92	Spanned ($p > 0.05$)
PyTorch LSTM LS	-0.36%	-0.06	39.0%	-0.427	-7.14	+0.486	+8.66	Spanned ($\alpha \approx 0$)
TFDMGA LS	-0.18%	-0.03	41.2%	-0.448	-7.84	+0.512	+9.12	Spanned ($\alpha \approx 0$)

and ElasticNet), net returns after 10 bps transaction fees suffer severe turnover decay, producing statistically significant negative net alphas ($\hat{\alpha} \approx -16.5\%$, $t \approx -2.01$, $p < 0.05$). These linear models are unspanned in a negative sense, generating net underperformance net of trading frictions. By contrast, deep sequence models (PyTorch LSTM and TFDMGA) effectively manage execution drag, yielding net strategy alphas that are statistically indistinguishable from zero ($\hat{\alpha} = -0.18\%$, $t = -0.03$, $p = 0.976$). The 5-factor model accounts for 41.2% of TFDMGA's return variance, with positive factor loadings on operating profitability (RMW beta = +0.512, $t = +9.12$). This confirms that deep sequence architectures function as dynamic factor allocation engines rather than discoverers of unpriced market arbitrage (Fama and French, 2015; Kelly et al., 2019).

5. Conclusion & Future Extensions

5.1 Summary of Research Findings

In this thesis, I conducted an empirical investigation into whether statistical machine learning and deep learning models improve cross-sectional stock return prediction across S&P 500 constituent equities over the ten-year period from 2015 through 2024 (Gu et al., 2020; Chen et al., 2024). By constructing a point-in-time aligned panel free from lookahead and survivorship bias (McLean and Pontiff, 2016), evaluating models across 5 expanding-window walk-forward test folds (2020–2024) (López de Prado, 2018), accounting for 10 bps transaction cost drag (Novy-Marx and Velikov, 2016), and estimating Fama-French five-factor spanning regressions (Fama and French, 2015; Carhart, 1997), my study provides answers to all four core research questions:

1. **Statistical Performance of Deep Sequence Encoders (RQ1):** Nonlinear sequence-based deep learning models outperform traditional linear regressions (Fama-MacBeth OLS, LASSO) and static tree classifiers (Random Forest, XGBoost). The proposed **TFDMGA network achieves high out-of-sample predictive accuracy**, yielding a daily Information Coefficient of $IC = +0.0348$, an annualized $ICIR = 3.12$, a ROC AUC of 0.6120, and a directional accuracy of 56.84%. A Diebold and Mariano (1995) test with Harvey et al. (1997) small-sample corrections indicates statistical significance over standard LSTM models ($DM = 2.41, p = 0.016$).
2. **Value of Multi-Modal Feature Fusion (RQ2):** Combining multi-frequency feature modalities (16 Technical, 30 Fundamental, 2 Sentiment, 5 Macro = 53 ML features) through a directed ring attention cascade (Technical $\leftarrow$ Sentiment $\leftarrow$ Fundamental $\leftarrow$ Macro) (Vaswani et al., 2017) and dynamic macro gating (Lim et al., 2021) produces higher out-of-sample signal stability compared to single-modality models (Cong et al., 2021). Sentiment indicators and price momentum

drive high-frequency signal directionality, while quarterly accounting quality provides structural drawdown defense.

3. **Trading Performance & Risk Overlays (RQ3):** Under 10 bps transaction costs and 1-day lagged execution (Novy-Marx and Velikov, 2016; Avramov et al., 2023), an initial **\$1,000 USD deposit** (Jan 2020) grows to **\$3,120.99 USD** under the baseline LSTM Q_5 long strategy. When paired with a 2:1 Take-Profit/Stop-Loss (TPSL) dynamic risk management overlay, drawdowns during macro stress periods (2020 COVID market decline, 2022 Fed rate hikes) are reduced, allowing capital to compound to **\$4,368.50 USD** for LSTM and **\$6,482.10 USD** for TFDMGA.
4. **Econometric Factor Spanning & Market Efficiency (RQ4):** Fama-French 5-factor spanning regressions yield an estimated net strategy alpha of $\hat{\alpha} = -0.18\%$ per year ($p = 0.976$, $R^2 = 41.2\%$) (Fama and French, 2015; Newey and West, 1987). Because net alpha is statistically indistinguishable from zero, 100% of the strategy's return generation is accounted for by systematic factor exposures ($Mkt - RF$, SMB , HML , RMW , CMA). This suggests that machine learning models function as **dynamic factor allocation engines** (Kelly et al., 2019; Kozak et al., 2020) rather than discoverers of unpriced market arbitrage, maintaining consistency with market efficiency (Fama, 1970).

5.2 Theoretical and Practical Implications

The empirical conclusions of this study carry several implications for quantitative finance practitioners and academic researchers:

- **For Empirical Asset Pricing:** The collapse of LASSO under linear MSE loss highlights the importance of rank-based loss functions ($\mathcal{L}_{\text{rank}}$, $\mathcal{L}_{\text{IC}}$) when training machine learning architectures on noisy financial return targets (Tibshirani, 1996; Kozak et al., 2020; Gu et al., 2020).
- **For Institutional Asset Managers:** Daily alpha signals experience turnover drag ($\approx 13\%$ daily turnover). Incorporating dynamic risk management rules (such as 2:1

TPSL circuit breakers) is essential for preserving capital and converting statistical predictions into tradeable wealth compounding (Novy-Marx and Velikov, 2016; Frazzini and Pedersen, 2014).

- **For Market Efficiency Debates:** Demonstrating that deep learning model returns are spanned by systematic Fama-French factors provides insight into why ML strategies generate positive returns. Machine learning does not violate market efficiency (Fama, 1970); rather, it automates dynamic factor rotation across macroeconomic regimes (Kelly et al., 2019; Chen et al., 2024).

5.3 Limitations of the Study

While this thesis adheres to standard backtesting protocols (López de Prado, 2018), certain empirical limitations exist:

1. **Execution Slippage & Market Impact:** Transaction costs were modeled using fixed round-trip friction levels (5, 10, 20 bps). Real-world execution of large institutional orders introduces nonlinear square-root market impact drag (Novy-Marx and Velikov, 2016).
2. **Universe Limitation:** The empirical panel is restricted to U.S. Large-Cap S&P 500 equities. Smaller, less liquid mid-cap or small-cap stocks may present different return patterns along with higher transaction costs (Avramov et al., 2023; Hou et al., 2020).

5.4 Future Research Extensions

The empirical baseline established in this thesis provides several avenues for future research extensions:

Extension 1: Spatio-Temporal Graph Neural Networks – Extending the stock sequence encoders in TFDMDGA to a Spatio-Temporal Graph Attention Network

(GAT), modeling stocks as graph nodes with dynamic edge weights $A_{ij,t}$ to capture cross-sectional supply chain linkages, industry spillovers, and dynamic return correlations (Vaswani et al., 2017; Chen et al., 2024).

Extension 2: Macro-Conditioned Hyper-LoRA Attention – Conditioning Transformer attention projections (W_Q, W_K, W_V) on macroeconomic regimes using Hypernetworks and Low-Rank Adaptation (LoRA) to allow dynamic adjustment to changing macroeconomic environments (Lim et al., 2021).

Extension 3: Continuous Deep Reinforcement Learning with Differentiable QP Layers – Replacing decile portfolio sorting with a continuous Deep Reinforcement Learning (DRL) agent (PPO) coupled with a differentiable Quadratic Programming solver layer (`cvxpylayers`) to optimize continuous portfolio weights directly against nonlinear Sharpe ratio rewards (López de Prado, 2020; Novy-Marx and Velikov, 2016).

Extension 4: Conditional Variational Autoencoder Factor Extraction – Developing a probabilistic Conditional Variational Autoencoder (CVAE) to extract nonlinear latent factor spaces from high dimensional characteristics (Gu et al., 2021; Kelly et al., 2019; Chen et al., 2024).

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